Nonparametric Tests: When the Assumptions Break Down

Not covered in textbook

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Learning Objectives

By the end of today's lecture, you will be able to:

1. Explain when and why nonparametric tests are preferred over parametric alternatives
2. Describe the core idea behind rank-based tests
3. Apply the Wilcoxon signed-rank test to paired or single-sample data
4. Apply the Wilcoxon rank-sum test to two independent samples
5. Apply the Kruskal-Wallis test to compare three or more groups
6. Choose the appropriate nonparametric test for a given study design

Roadmap for Today

Part 1: Parametric vs. Nonparametric

- When do the assumptions break down?
- The rank-based approach
- What we gain and what we lose
- The complete comparison table

Part 2: Wilcoxon Signed-Rank Test

- The sign test (brief)
- Wilcoxon signed-rank: the upgrade
- Glaucoma example (paired data)
- Running it in R

Part 3: Wilcoxon Rank-Sum Test

- The nonparametric two-sample t-test
- Rat neuron example
- Running it in R

Part 4: Kruskal-Wallis Test

- The nonparametric ANOVA
- IL-6 example
- Running it in R
- Post-hoc follow-up

Part 1: Parametric vs. Nonparametric

What do all our tests so far have in common?

Every test we've learned assumes the data come from a specific distribution:

Test	Distributional assumption
One-sample t-test	Normal population (or large n)
Paired t-test	Normal differences (or large n)
Two-sample t-test	Normal populations (or large n)
ANOVA	Normal within each group (or large n)

These are called *parametric* tests — they assume the population can be described by specific parameters (μ, σ) under a named distribution.

What if the assumptions don't hold?

Small samples + non-normal data + outliers = parametric tests may not be valid. Nonparametric tests are the answer.

When do parametric assumptions break down?

Situation 1: Small samples with skewed data

- $n = 8$ per group, right-skewed outcome
- Central limit theorem doesn't save you here
- t-test p-values may not be reliable

Situation 2: Outliers that aren't errors

- Biological data often has extreme values
- A single outlier can dominate the mean
- The outlier is real — you can't just remove it

Situation 3: Ordinal outcomes

- Pain scale (1–10), severity scores, Likert items
- Means aren't always meaningful
- "Average pain of 4.7" — what does that mean?

Situation 4: Heavy-tailed distributions

- Gene expression, antibody titers, viral loads
- Extreme values are common, not aberrant
- Log-transformation helps, but isn't always appropriate

The rank-based approach

Key idea: Instead of testing raw values, convert them to ranks first.

Step 1: Take your original data

Patient	Viral load
A	125
B	3024
C	780
D	55
E	426

Step 2: Replace values with ranks

Patient	Viral load	Rank
D	55	1
A	125	2
E	426	3
C	780	4
B	3024	5

Why ranks work

Patient B's viral load of 3024 is extreme, but as a rank it's just "5" — the highest. Ranks compress outliers automatically. The rank-sum test uses these ranks instead of raw values, so no distributional assumption is needed.

Pros and cons of nonparametric tests

Advantages

- **Fewer assumptions** — no normality required
- **Robust to outliers** — extreme values become just another rank
- **Work with small samples** — exact p-values available
- **Can handle ordinal data** — ranks are meaningful for ordered categories

Trade-offs

- **Less powerful** — converting to ranks discards information about magnitude
- If parametric assumptions *are* met, the t-test or ANOVA will detect effects more efficiently
- Confidence intervals are harder to compute
- Ties in ranks require corrections

The bottom line

Use parametric tests when assumptions are met — they're more powerful. Switch to nonparametric when you have small samples, severe skew, heavy outliers, or ordinal data.

The complete comparison table

Study design	Parametric test	Nonparametric alternative
Single sample, numeric	One-sample t-test	Wilcoxon signed-rank
Paired sample, numeric	Paired t-test	Wilcoxon signed-rank
Two independent samples, numeric	Two-sample t-test	Wilcoxon rank-sum
3+ independent samples, numeric	ANOVA	Kruskal-Wallis

What about categorical outcomes?

For 2×2 tables with small expected cell counts, we use **Fisher's exact test** — we covered this already in Lesson 14. It's the nonparametric alternative to the chi-squared test for categorical data.

Part 2: Wilcoxon Signed-Rank Test

First: the sign test (brief)

The **sign test** is the simplest nonparametric test for paired or single-sample data.

Paired setting: each subject i has two measurements — a first observation x_{i1} (e.g., pre-treatment) and a second observation x_{i2} (e.g., post-treatment). Compute the within-subject difference:

$$d_i = x_{i2} - x_{i1}$$

Single-sample setting: each subject has one observation x_i, and we compare it against a hypothesized value μ_0:

$$d_i = x_i - \mu_0$$

In both cases, the test proceeds the same way:

- Count the number of **positive** differences (D^+) and **negative** differences (D^-)
- Ignore ties ($d_i = 0$)
- Under H_0: differences are symmetric around 0, so $D^+ \sim \text{Binomial}(n, 0.5)$

Why we don't use the sign test

The problem with the sign test

The sign test only uses the **direction** of each difference — not the **size**.

A difference of 0.01 and a difference of 10.0 both count as a single "+". This throws away most of the information in your data and makes the test much less powerful than it could be.

The Wilcoxon signed-rank test solves this by incorporating the magnitude of differences through ranking.

The Wilcoxon signed-rank test

The upgrade over the sign test: We rank the *absolute values* of the differences, then use the signs.

Conceptual steps:

1. Compute differences d_i (drop any zeros)
2. Rank the **absolute values** $|d_i|$ from smallest to largest (ties get average ranks)
3. Attach the original sign (+/−) back to each rank
4. Sum the positive ranks (T^+) and negative ranks (T^-)
5. If H_0 is true, T^+ and T^- should be roughly equal

What's being tested

H_0: The population of differences is **symmetric around zero**

H_A: The population of differences is **not symmetric around zero**

This is equivalent to testing whether the median difference is zero — provided the distribution is symmetric.

The glaucoma example

Intraocular pressure in glaucoma patients is often reduced by treatment with adrenaline. A **new synthetic drug** is being evaluated — more expensive, but potentially more effective.

- 7 patients treated with **both** drugs (one eye each)
- Outcome: reduction in pressure after treatment
- Larger values = greater pressure reduction

Question: Do the two drugs produce the same reduction in pressure?

```r
IOP
```

```
# A tibble: 7 × 4
  patient adrenaline synthetic      d
    <int>      <dbl>     <dbl>  <dbl>
1       1        3.5       3.2 -0.300
2       2        2.6       3.1  0.5
3       3        3         3.3  0.300
4       4        1.9       2.4  0.5
5       5        2.9       2.9  0
6       6        2.4       2.8  0.4
7       7        2         2.6  0.6
```

Step 1: State hypotheses

H_0: The differences in intraocular pressure reduction (synthetic – adrenaline) are symmetrically distributed around zero

H_A: The differences are not symmetrically distributed around zero

In plain language:

H_0: The two drugs produce the same median reduction in intraocular pressure

H_A: The two drugs differ in median reduction of intraocular pressure

Step 2: Set significance level

$$\alpha = 0.05$$

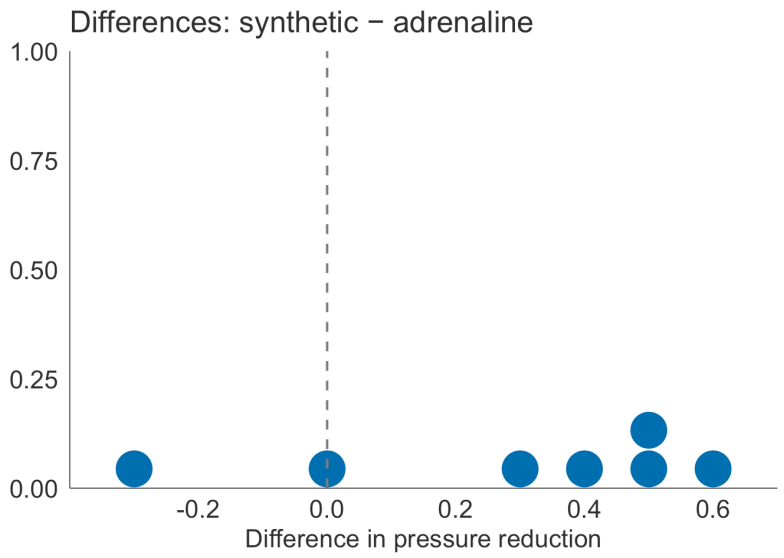
Step 3: Check assumptions

Wilcoxon signed-rank assumptions:

1. Paired observations (✓ — same patients, one eye each)
2. Differences are **continuous** (✓ — pressure measurements)
3. Distribution of differences is **symmetric** around the median (check visually)

- Assumptions 1 and 2 are verified from the study design alone.
- For assumption 3, we plot the distribution of differences d_i and look for approximate symmetry around zero — no need for perfect normality, just no severe skew or outliers pulling in one direction.

Step 3: Check assumptions — visually



What to look for: approximate symmetry around zero — no severe skew, no outliers pulling strongly in one direction.

What we see:

- Most differences are positive — the synthetic drug tends to reduce pressure more
- The spread on both sides of zero is roughly balanced
- With a small sample, we do not assume normality — only approximate symmetry

✓ Assumption 3 is satisfied.

How ranking works — conceptually

Patient	Adrenaline	Synthetic	d	abs(d)	Rank of abs(d)	Signed rank
1	3.5	3.2	-0.3	0.3	1.5	**-1.5**
2	2.6	3.1	0.5	0.5	4.5	**4.5**
3	3.0	3.3	0.3	0.3	1.5	**1.5**
4	1.9	2.4	0.5	0.5	4.5	**4.5**
6	2.4	2.8	0.4	0.4	3.0	**3.0**
7	2.0	2.6	0.6	0.6	6.0	**6.0**

- Patient 5 had $d = 0$ → dropped from analysis (effective $n = 6$)
- $T^+ = 1.5 + 3 + 4.5 + 4.5 + 6 = 19.5$ (sum of positive ranks)
- $T^- = 1.5$ (sum of negative ranks, in absolute value)
- If H_0 true: T^+ and T^- should be roughly equal

Steps 4–5: Test statistic and p-value in R

```r
# exact = FALSE because ties exist in the nonzero differences
sr_result <- wilcox.test(IOP$d, mu = 0,
                         alternative = "two.sided",
                         exact = FALSE)

sr_result %>%
  broom::tidy()
```

```
# A tibble: 1 × 4
  statistic p.value method                                                alternative
      <dbl>   <dbl> <chr>                                                 <chr>
1      19.5  0.0731 Wilcoxon signed rank test with continuity corre… two.sided
```

Reading the output

- `statistic` = T^+ = 19.50 (sum of positive ranks)
- `p.value` = p-value using normal approximation — R uses the exact method only when sample is small **and** there are no ties. Because our nonzero differences contain ties (0.3 appears twice, 0.5 appears twice), R falls back to the normal approximation automatically. Setting `exact = FALSE` makes this explicit and prevents a warning.

Using rstatix

```r
library(rstatix)

IOP %>%
  wilcox_test(d ~ 1, mu = 0,
              alternative = "two.sided",
              exact = FALSE)
```

```
# A tibble: 1 × 6
  .y.   group1 group2         n statistic      p
* <chr> <chr>  <chr>      <int>     <dbl>  <dbl>
1 d     1      null model     7      19.5 0.0731
```

Step 6: Conclusion

With p = 0.073 and $\alpha = 0.05$, we **fail to reject** H_0.

Conclusion

There is insufficient evidence that the median difference in intraocular pressure reduction between the synthetic drug and adrenaline differs from zero (Wilcoxon signed-rank test, T^+ = 19.50, p = 0.073).

The synthetic drug showed a numerically higher median reduction (difference = 0.40 mmHg), but this difference was not statistically significant at $\alpha = 0.05$ with only 6 evaluable patients. A larger study would be needed to draw a definitive conclusion.

Note on sample size

$n = 7$ (effective $n = 6$) gives this study very low power. Failing to reject H_0 here does not mean the drugs are equivalent — it may simply mean we don't have enough data.

Part 3: Wilcoxon Rank-Sum Test

The nonparametric two-sample t-test

When the two-sample t-test assumptions don't hold and the groups are **independent**, use the **Wilcoxon rank-sum test** (also called the Mann-Whitney U test).

The idea:

1. Combine both groups into one dataset
2. Rank all observations from 1 to N
3. Sum the ranks within each group
4. If the groups have the same median, the rank sums should be roughly equal
5. If one group tends to have higher ranks, that group has a higher median

Hypotheses

H_0: The distributions of the two populations are identical

H_A: The distributions of the two populations are not identical

(When both groups share the same shape, any difference reflects a shift in location — i.e., a difference in median.)

The rat neuron example

A study measured **evoked membrane current** of dental sensory neurons in rats under two conditions:

- **Control:** standard conditions
- **Treatment:** capsaicin + capsazepine (CPZ)

6 rats per group. Membrane current values (pA) are highly right-skewed with a wide range — a classic case for the rank-sum test.

Question: Does CPZ treatment alter the median membrane current?

Rat neuron data

```r
CPZ
```

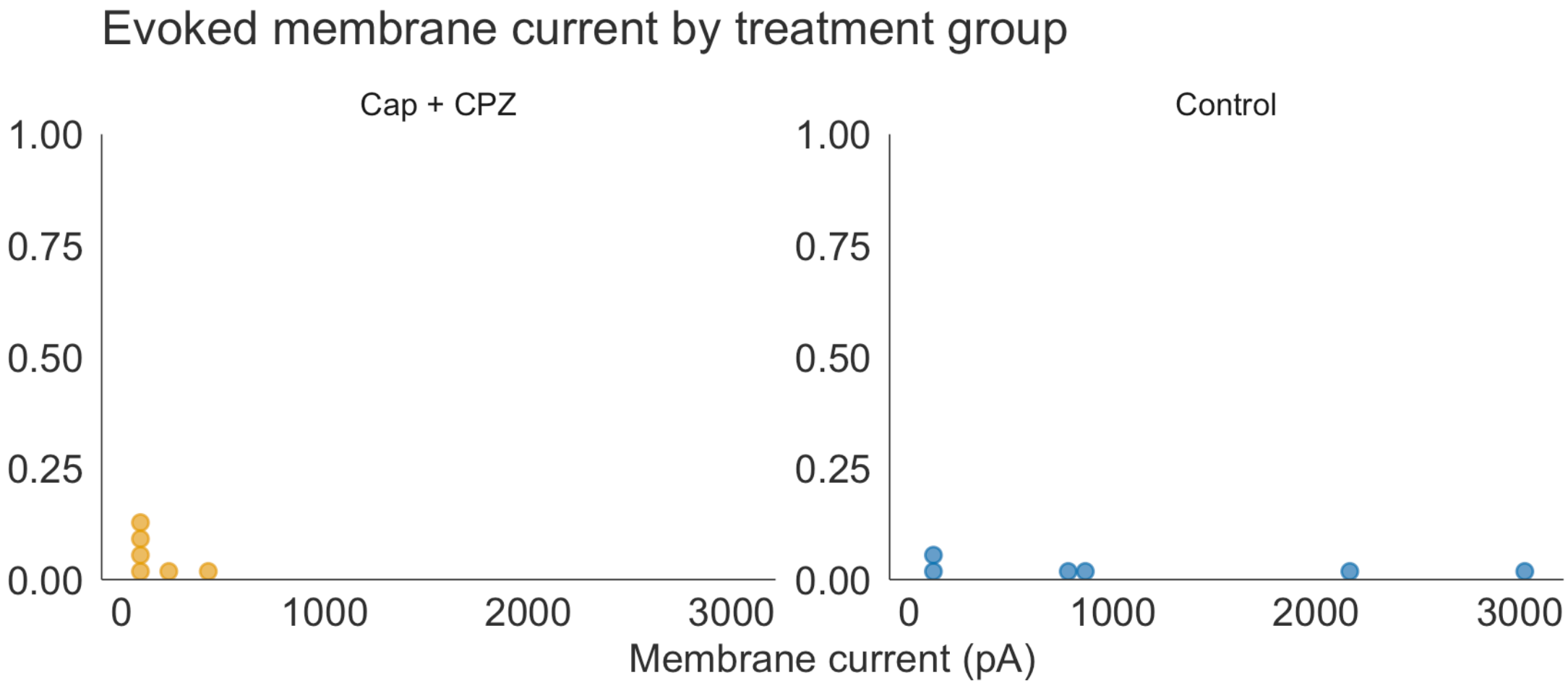
```
# A tibble: 12 × 2
   group     current
   <chr>       <dbl>
 1 Control      3024
 2 Control      2164
 3 Control       864
 4 Control       780
 5 Control       125
 6 Control       110
 7 Cap + CPZ     426
 8 Cap + CPZ     232
 9 Cap + CPZ     130
10 Cap + CPZ      94
11 Cap + CPZ      75
12 Cap + CPZ      55
```

```r
CPZ %>%
  group_by(group) %>%
  summarise(
    n      = n(),
    median = median(current),
    mean   = round(mean(current), 0)
  )
```

```
# A tibble: 2 × 4
  group         n median  mean
  <chr>     <int>  <dbl> <dbl>
1 Cap + CPZ     6    112   169
2 Control       6    822  1178
```

Visualizing the data

▶ Code



Both groups are **right-skewed** with similar shape — consistent with the rank-sum assumption. The control group has a much higher median.

How ranking works — conceptually

```r
CPZ %>%
  arrange(current) %>%
  mutate(rank = rank(current,
                     ties.method = "average"))
```

```
# A tibble: 12 × 3
   group     current  rank
   <chr>       <dbl> <dbl>
 1 Cap + CPZ      55     1
 2 Cap + CPZ      75     2
 3 Cap + CPZ      94     3
 4 Control       110     4
 5 Control       125     5
 6 Cap + CPZ     130     6
 7 Cap + CPZ     232     7
 8 Cap + CPZ     426     8
 9 Control       780     9
10 Control       864    10
11 Control      2164    11
12 Control      3024    12
```

- $W_{\text{Control}} = 4 + 5 + 9 + 10 + 11 + 12 = 51$
- $W_{\text{Cap+CPZ}} = 1 + 2 + 3 + 6 + 7 + 8 = 27$

- If H_0 true, we'd expect rank sums to be roughly equal (both ≈ 39).

Step 1: State hypotheses

H_0: The distribution of evoked membrane current is the same in the control and CPZ-treated populations

H_A: The distributions of evoked membrane current in the control and CPZ-treated populations are not identical

Steps 2–3: Significance level and assumptions

$\alpha = 0.05$

Wilcoxon rank-sum assumptions:

1. Two independent samples (✓ — different rats in each group)
2. Continuous outcome (✓ — membrane current in pA)
3. Both distributions have the **same shape** — check visually

The dotplots showed both groups are right-skewed with similar spread. ✓

Steps 4–5: Test statistic and p-value in R

```r
# exact = TRUE is appropriate: both n < 10, no ties in this dataset
rs_result <- wilcox.test(current ~ group, data = CPZ,
                         alternative = "two.sided",
                         exact = TRUE)

rs_result %>%
  broom::tidy()
```

```
# A tibble: 1 × 4
  statistic p.value method                      alternative
      <dbl>   <dbl> <chr>                       <chr>
1         6  0.0649 Wilcoxon rank sum exact test two.sided
```

Reading the output

- `statistic` = W = rank sum for the first alphabetical group
- `p.value` = exact p-value (appropriate here: both $n < 10$, no ties)
- For larger samples or when ties exist, R automatically uses the normal approximation

Using `rstatix`

```r
CPZ %>%
  wilcox_test(current ~ group,
              alternative = "two.sided",
              exact = TRUE)
```

```
# A tibble: 1 × 7
  .y.     group1    group2     n1    n2 statistic      p
* <chr>   <chr>     <chr>   <int> <int>     <dbl>  <dbl>
1 current Cap + CPZ Control     6     6         6 0.0649
```

Step 6: Conclusion

With p = 0.065 and $\alpha = 0.05$, we **fail to reject** H_0.

Conclusion

There is suggestive but inconclusive evidence that the distributions of evoked membrane current differ between the control group and the group exposed to capsaicin + CPZ (Wilcoxon rank-sum test, W = 6.00, p = 0.065).

The control group had a substantially higher median current (median = 822 pA) compared to the CPZ group (median = 112 pA), but the small sample size ($n = 6$ per group) limits statistical power.

Part 4: Kruskal-Wallis Test

The nonparametric ANOVA

When ANOVA assumptions fail with 3 or more groups, use the **Kruskal-Wallis test**.

It extends the rank-sum test to k groups:

1. Combine all groups, rank all observations 1 to N
2. Compute the sum of ranks within each group: $R_1, R_2, \ldots, R_k$
3. If H_0 is true, rank sums should be similar across groups
4. The test statistic H quantifies how different the rank sums are

$$H_0 : \text{The distributions of all } k \text{ groups are identical}$$

$$H_A : \text{At least one group has a different distribution}$$

Tip

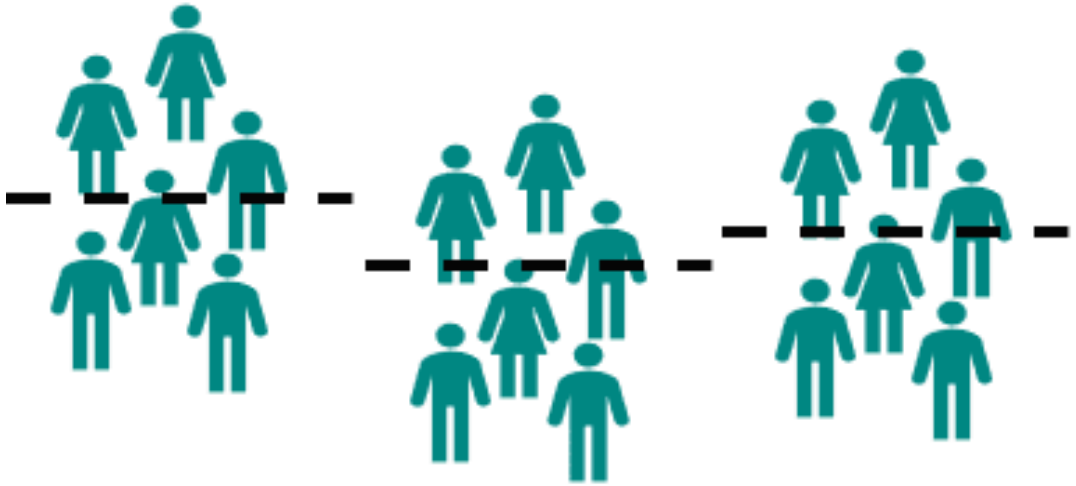
The K-W test statistic H follows a χ^2 distribution with $k - 1$ degrees of freedom. When $k = 2$, K-W gives the same result as the rank-sum test.

When all groups share the same distributional shape, any difference reflects a shift in location — i.e., a difference in median.

Kruskal-Wallis vs ANOVA

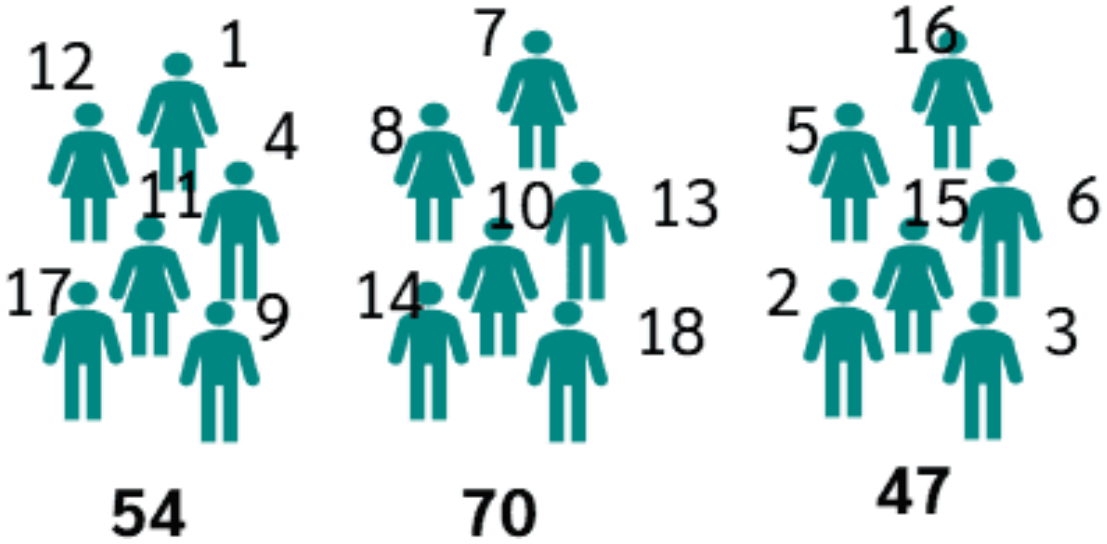
Analysis of variance

Is there a difference in mean?



Kruskal-Wallis-Test

Is there a difference in the rank totals?



Source: numiqo Team (2026). numiqo: Online Statistics Calculator. https://numiqo.com

The IL-6 example

Serum **interleukin-6 (IL-6)** is an inflammatory cytokine commonly elevated in autoimmune and infectious disease. A researcher measured IL-6 levels (pg/mL) in patients classified into three disease severity groups.

- **Mild** ($n = 8$), **Moderate** ($n = 8$), **Severe** ($n = 8$)
- IL-6 values are highly right-skewed — a classic case for the Kruskal-Wallis test

Question: Do IL-6 levels differ across disease severity groups?

IL-6 data

```r
il6_data
```

```
# A tibble: 24 × 2
   severity   il6
   <fct>    <dbl>
 1 Mild       3.1
 2 Mild       3.2
 3 Mild       3.8
 4 Mild       3.7
 5 Mild       3.1
 6 Mild       2.8
 7 Mild       5.4
 8 Mild       4.2
 9 Moderate  23
10 Moderate   8.8
# ℹ 14 more rows
```

```r
il6_data %>%
  group_by(severity) %>%
  summarise(
    n      = n(),
    median = median(il6),
    mean   = mean(il6),
    sd     = sd(il6)
  )
```

```
# A tibble: 3 × 5
  severity     n median  mean     sd
  <fct>    <int>  <dbl> <dbl>  <dbl>
1 Mild         8   3.45  3.66  0.838
2 Moderate     8  14.8  16.3   7.57
3 Severe       8  68.2  86.4  42.6
```

Why not ANOVA?

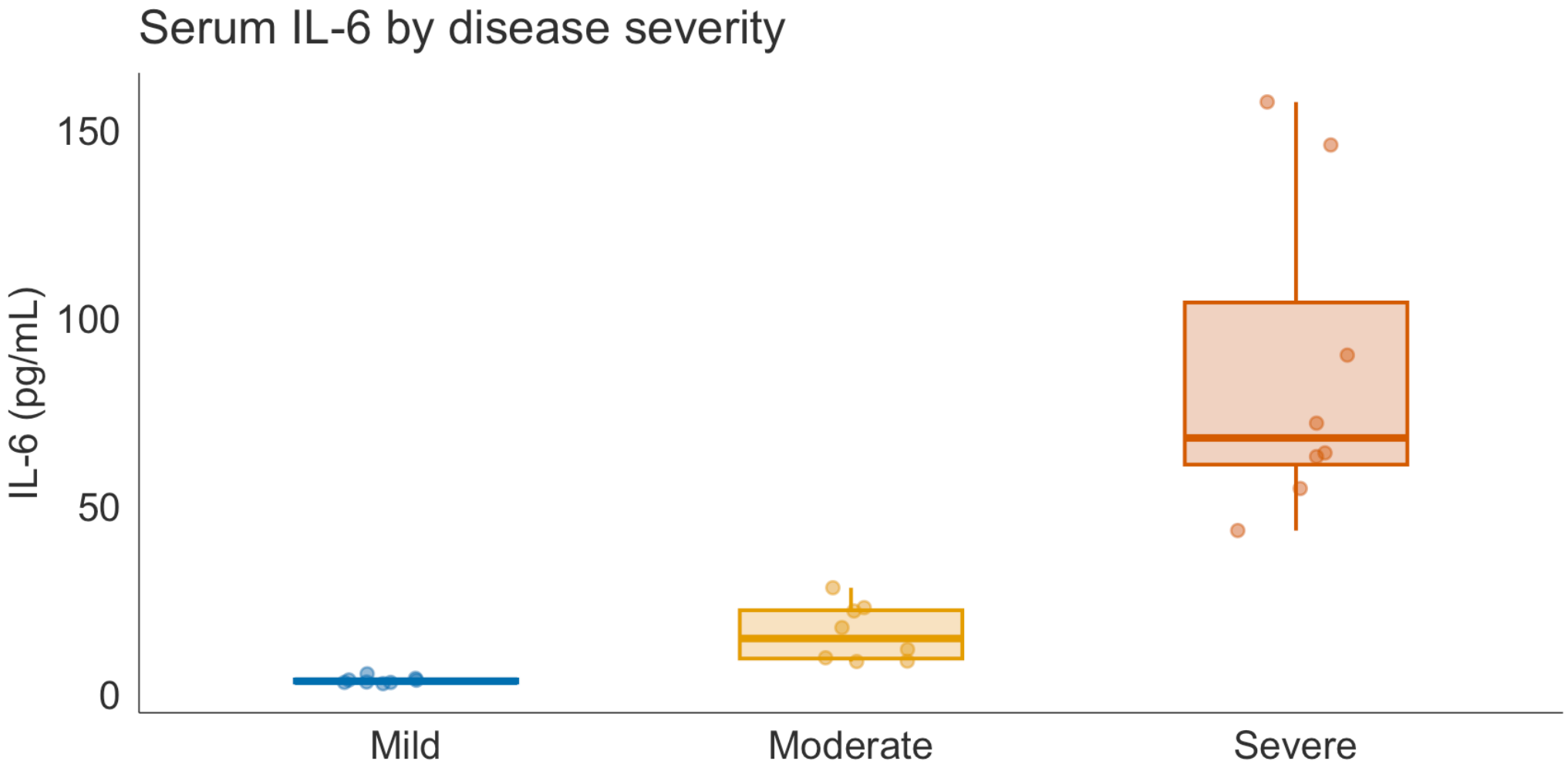
```r
# Check the equal-variance assumption
il6_data %>%
  group_by(severity) %>%
  summarise(sd = sd(il6)) %>%
  summarise(ratio = max(sd) / min(sd))
```

```
# A tibble: 1 × 1
  ratio
  <dbl>
1  50.8
```

SD ratio > 2 and groups are right-skewed → ANOVA assumptions violated. Kruskal-Wallis is the right choice.

Visualizing IL-6 by severity group

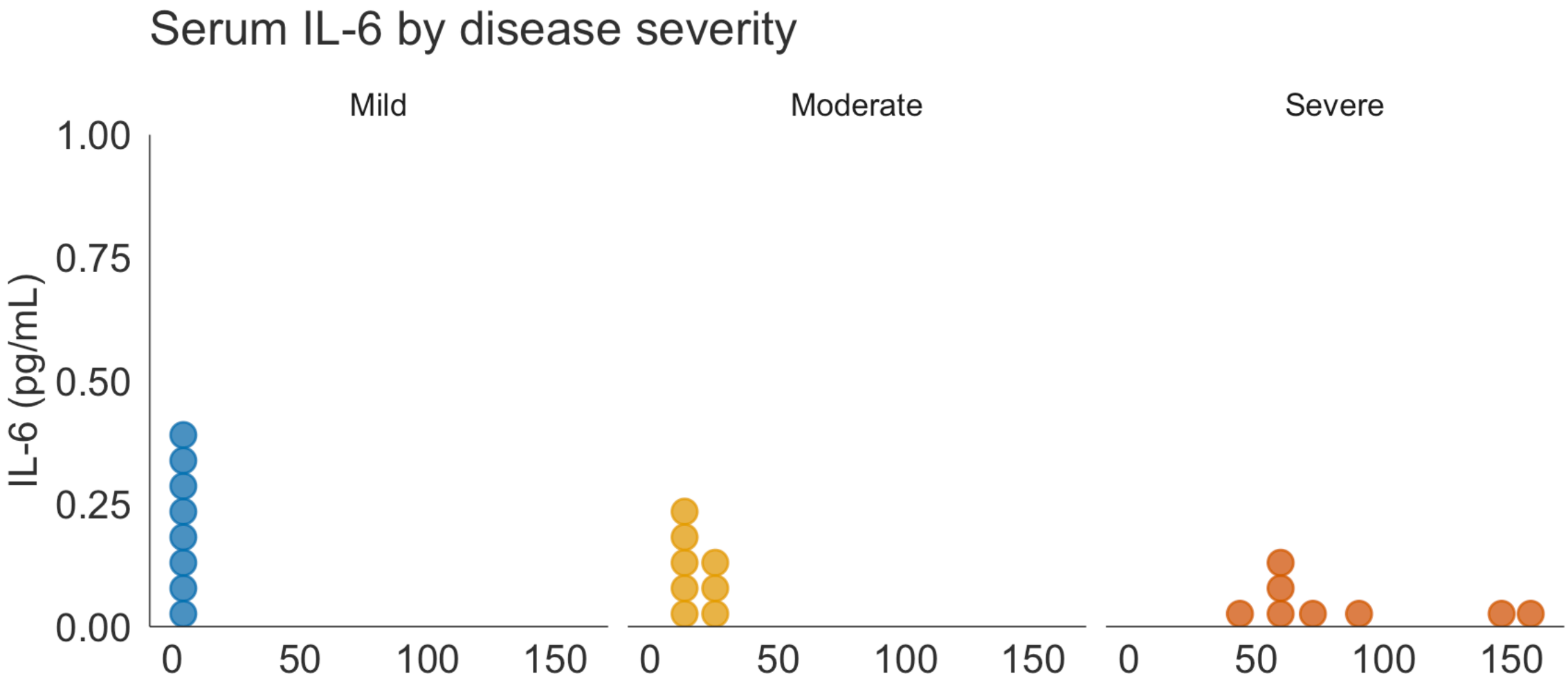
▶ Code



Right-skewed distributions across all groups. Median IL-6 appears to increase with severity. Groups are clearly not normally distributed.

Dotplot of IL-6 by severity group

▶ Code



Right-skewed distributions within each group — consistent with what the boxplots showed.

Step 1: State hypotheses

H_0: The distributions of serum IL-6 are identical across all three severity groups

H_A: At least one severity group has a different distribution of serum IL-6

Steps 2–3: Significance level and assumptions

$\alpha = 0.05$

Kruskal-Wallis assumptions:

1. Independent observations across groups (✓ — different patients in each group)
2. Continuous or ordinal outcome (✓ — IL-6 in pg/mL)
3. All groups come from populations with the **same distributional shape** — check visually

The previous plots suggest right-skewed distributions across all three groups. ✓

Steps 4–5: Test statistic and p-value in R

```r
kw_result <- kruskal.test(il6 ~ severity, data = il6_data)

kw_result %>%
  broom::tidy()
```

```
# A tibble: 1 × 4
  statistic   p.value parameter method
      <dbl>     <dbl>     <int> <chr>
1      20.5 0.0000356         2 Kruskal-Wallis rank sum test
```

Reading the output

- `statistic` = H = Kruskal-Wallis test statistic
- `p.value` = p-value from χ^2 distribution with df = $k - 1$ = 2
- `parameter` = degrees of freedom

Using `rstatix`

```r
il6_data %>%
  kruskal_test(il6 ~ severity)
```

```
# A tibble: 1 × 6
  .y.       n statistic    df         p method
* <chr> <int>     <dbl> <int>     <dbl> <chr>
1 il6      24      20.5     2 0.0000356 Kruskal-Wallis
```

Step 6: Conclusion

With H = 20.49, df = 2, p = < 0.001, we **reject** H_0.

Conclusion

There is strong evidence that the distribution of serum IL-6 differs across at least two disease severity groups (Kruskal-Wallis test, H = 20.49, df = 2, p = < 0.001).

Descriptively, median IL-6 increased with severity — mild (median = 3.4 pg/mL), moderate (median = 14.8 pg/mL), and severe (median = 68.2 pg/mL).

Post-hoc comparisons after K-W (1/2)

Just like ANOVA, a significant K-W test tells us *something* differs — not *which* pairs. Follow up with pairwise Wilcoxon rank-sum tests with a correction:

```r
pairwise.wilcox.test(il6_data$il6, il6_data$severity,
                     p.adjust.method = "bonferroni")
```

```
    Pairwise comparisons using Wilcoxon rank sum test with continuity correction

data:  il6_data$il6 and il6_data$severity

         Mild    Moderate
Moderate 0.00279 -
Severe   0.00279 0.00047

P value adjustment method: bonferroni
```

Post-hoc comparisons after K-W (2/2)

Alternative using `rstatix`:

```r
il6_data %>%
  wilcox_test(il6 ~ severity,
              p.adjust.method = "bonferroni",
              paired = FALSE)
```

```
# A tibble: 3 × 9
  .y.   group1   group2      n1    n2 statistic        p    p.adj p.adj.signif
* <chr> <chr>    <chr>    <int> <int>     <dbl>    <dbl>    <dbl> <chr>
1 il6   Mild     Moderate     8     8         0 0.000931 0.003    **
2 il6   Mild     Severe       8     8         0 0.000931 0.003    **
3 il6   Moderate Severe       8     8         0 0.000155 0.000465 ***
```

Tip

There is no nonparametric equivalent of Tukey HSD. Bonferroni correction is a straightforward choice — it controls the family-wise error rate across all pairwise comparisons. With only 3 pairs here, the penalty for using Bonferroni is small.

Wrap-Up

Choosing the right nonparametric test

Question	Test
Is the median different from a value? (single sample)	Wilcoxon signed-rank
Do paired measurements differ?	Wilcoxon signed-rank
Do two independent groups differ?	Wilcoxon rank-sum
Do 3+ independent groups differ?	Kruskal-Wallis

Key decision rule:

1. Are parametric assumptions met? → Use parametric test (more power)
2. Small n, skewed data, outliers, ordinal outcome? → Use nonparametric test

R functions summary:

```r
# Signed-rank (paired / one-sample)
wilcox.test(data$d, mu = 0)

# Rank-sum (two independent groups)
wilcox.test(outcome ~ group,
            data = data)

# Kruskal-Wallis (3+ groups)
kruskal.test(outcome ~ group,
             data = data)

# Post-hoc after K-W
pairwise.wilcox.test(
  data$outcome, data$group,
  p.adjust.method = "bonferroni"
  )
```

Connecting to what's next

Lesson 18 (Wednesday): Correlation and Simple Linear Regression

- Both variables are continuous — we're no longer comparing groups
- Pearson correlation: the parametric approach
- Spearman correlation: the nonparametric analogue (ranks again!)
- Simple linear regression: predicting one variable from another

The big picture

Nonparametric tests are a safety net — they work when parametric assumptions fail. But they're not magic. If assumptions are met, stick with t-tests and ANOVA. If they aren't, you now have the right tools to handle it.

The rank-based logic you saw today also underlies **Spearman correlation**, which we'll see on Wednesday — so this was good preparation.

References and Further Reading

Article

Nahm FS. **Nonparametric statistical tests for the continuous data: the basic concept and the practical use**. *Korean Journal of Anesthesiology*. 2016;69(1):8–14. doi:10.4097/kjae.2016.69.1.8

- Accessible overview of when and why to use nonparametric tests, with practical examples drawn from medical research

Web resources (numiqo Team, 2026. numiqo: Online Statistics Calculator. https://numiqo.com)

- **Parametric vs. nonparametric tests** — overview and decision guidance
- **Wilcoxon signed-rank test** — paired and single-sample
- **Mann-Whitney U / Wilcoxon rank-sum test** — two independent groups
- **Kruskal-Wallis test** — three or more groups